

Chapter 3. Analytic Geometry



⁰Equivalent to 2 slots. Lecturers can use extra materials for other slots in the chapter.

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- 1 3.1 Norm
- 2 3.2 Inner Products
- 3 3.3 Lengths and Distances
- 4 3.4 Angles and Orthogonality
- 5 3.5 Orthonormal Basis
- 6 3.6 Orthogonal Complement
- 7 3.8 Orthogonal Projections

Let V be a vector space.

Definition

A **norm** on V is a non-negative function

$$\|\cdot\| : V \rightarrow \mathbb{R}$$

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Example

Let p be a positive number. The l_p -norm $\|\cdot\|_p$ in $\mathbb{R}^n$:

$$\|\mathbf{x}\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad \text{for any } \mathbf{x} = [x_1 \ \cdots \ x_n]^T \in \mathbb{R}^n.$$

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Show that the mapping $\mathcal{B} : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

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Consider $V = \mathbb{R}^2$. Define

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 - x_1 y_2 + x_2 y_1 + x_2 y_2.$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product in $\mathbb{R}^2$.

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where $A = [A_{ij}] = [\langle \mathbf{b}_i, \mathbf{b}_j \rangle]$ and $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ are the coordinates of $\mathbf{x}$ and $\mathbf{y}$ w.r.t the basis B .

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Theorem

For an n -dimensional vector space V and an ordered basis B of V , it holds that $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $A \in \mathbb{R}^{n \times n}$ with

$$\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^T A \hat{\mathbf{y}}.$$

Definition

Consider an inner product space $(V, \langle \cdot, \cdot \rangle)$. Then

- The norm induced by the inner product $\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$

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Let the inner product $\langle \cdot, \cdot \rangle$ be defined on $\mathbb{R}^2$ by

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The **angle** between the vectors $\mathbf{x}$ and $\mathbf{y}$ is the number $\omega \in [0, \pi]$ defined by

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Consider an n -dimensional inner product vector space $(V, \langle \cdot, \cdot \rangle)$ and a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of V .

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B is called an **orthonormal basis (ONB)** if

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3.8 Orthogonal Projections

Let V be a vector space and $U \subseteq V$ a subspace of V .

Definition

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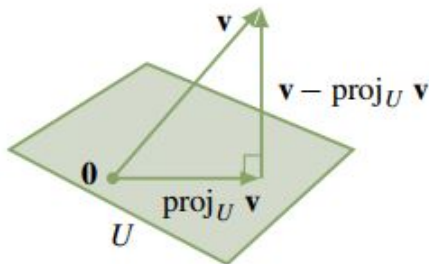
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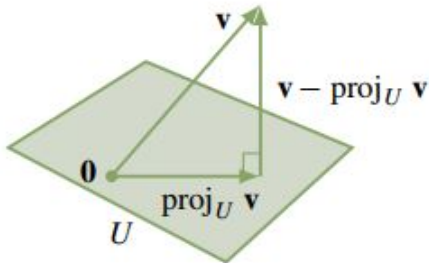


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Now we consider orthogonal projections of vectors $\mathbf{x} \in \mathbb{R}^n$ onto lower-dimensional subspaces $U \subseteq \mathbb{R}^n$ with a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$. It is easy to find $\text{proj}_U(\mathbf{x})$ after bringing B into a orthonormal basis by Gram-Schmidt process, then apply Projection Theorem. The following we are using a different approach.

$$\text{proj}_U(\mathbf{x}) = \sum_{i=1}^m \lambda_i \mathbf{b}_i = B\lambda,$$

$$B = [\mathbf{b}_1 \quad \cdots \quad \mathbf{b}_m], \quad \lambda = [\lambda_1 \quad \cdots \quad \lambda_m]^T$$

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